

FINITE CONGRUENCE COVERINGS FOR THE ERDŐS–STRAUS EQUATION AND STERILE PROFINITE POINTS

ANONYMOUS

ABSTRACT. ABSTRACT-PLACEHOLDER

1. INTRODUCTION

INTRO-PLACEHOLDER

2. COVERINGS AS CLOPEN CLASSES

2.1. The seven families. We use the families of Elsholtz and Tao [2, Prop. 1.9]. Each is indexed by a triple κ of positive integers subject to a side condition, and defines a set of residue classes:

family	parameters	side condition	class
I1	a, d, f	$f \mid 4a^2d + 1$	$n \equiv -f \pmod{4ad}$
I2	a, c, f	$(4ac, f) = 1$	$n \equiv -f \pmod{4ac}, n \equiv -c/a \pmod{f}$
I3	c, d, f	$(4cd, f) = 1$	$n \equiv -f \pmod{4cd}, n^2 \equiv -4c^2d \pmod{f}$
I4	a, b, e	$e \mid a + b, (e, 4ab) = 1$	$n \equiv -1/e \pmod{4ab}$
II1	a, b, e	$e \mid a + b, (e, 4ab) = 1$	$n \equiv -e \pmod{4ab}$
II2	a, d, f	$4ad \mid f + 1$	$n \equiv -4a^2d \pmod{f}$
II3	a, d, e	$(4ad, e) = 1$	$n \equiv -4a^2d - e \pmod{4ade}$

On its class each family carries explicit solutions. With $\pi^I(a, b, c, d) = (abdn, acd, bcd)$ and $\pi^{II}(a, b, c, d) = (abd, acdn, bcdn)$ [2, §2], the remaining coordinates are

I1	$e = (4a^2d + 1)/f, b = (ne + 1)/(4ad), c = (n + f)/(4ad)$
I2	$d = (n + f)/(4ac), b = (na + c)/f$
I3	$a = (n + f)/(4cd), b = (n + (n^2 + 4c^2d)/f)/(4cd)$
I4	$c = (a + b)/e, d = (ne + 1)/(4ab)$
II1	$c = (a + b)/e, d = (n + e)/(4ab)$
II2	$c = (f + 1)/(4ad), b = (nc + a)/f$
II3	$c = (n + 4a^2d + e)/(4ade), b = ce - a = (n + e)/(4ad)$

and the solution is π^I for I1–I4 and π^{II} for II1–II3.

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Lemma 2.1 (PROVED; [2, §§2, 10]). *Let κ be an admissible parameter triple of one of the seven families and let $n \geq 1$ be an integer in its class. Then the coordinates above are positive integers, and the resulting (x, y, z) is a solution of $4/n = 1/x + 1/y + 1/z$ in positive integers.*

Proof. The identity $4/n = 1/x + 1/y + 1/z$ follows from the defining relations of the varieties $\Sigma_n^I, \Sigma_n^{II}$ of [2, (2.1)–(2.9), (2.13)–(2.21)]. Integrality on the class is a direct check; for example in I3, $n \equiv -f \pmod{4cd}$ gives $a \in \mathbb{Z}$, the second congruence gives $Y := (n^2 + 4c^2d)/f \in \mathbb{Z}$, and $fY \equiv f^2 \pmod{4cd}$ with f invertible modulo $4cd$ gives $Y \equiv f \equiv -n$, so $b \in \mathbb{Z}$. With κ fixed, every coordinate is a polynomial in n with non-negative coefficients that is positive at $n = 1$, so all coordinates are positive for $n \geq 1$. $\square$

The positivity statement for all $n \geq 1$ (not only for large n) was checked in the project's review of [12]; the degrees of (x, y, z) in n are at most 4.

ET call a primitive residue class *solvable by polynomials* if there are polynomials P_1, P_2, P_3 with rational coefficients, taking positive integer values at all sufficiently large n of the class, with $4/n = \sum 1/P_i(n)$ identically. Lemma 2.1 says that every primitive class contained in one of the seven families is solvable by polynomials. We use the converse in the following form.

Cited theorem 2.2 ([2, Prop. 1.9]). *If a primitive residue class is solvable by polynomials, then all sufficiently large primes in it lie in the union of finitely many classes of the seven families.*

(ET state this separately for Type I and Type II solvable classes, with the families II1–II4 and II1–II3 respectively; every solvable primitive class is of one of the two types [2, p. 8].)

2.2. Profinite language. Let $\widehat{\mathbb{Z}} = \prod_q \mathbb{Z}_q$ be the profinite completion of $\mathbb{Z}$, with unit group $\widehat{\mathbb{Z}}^\times = \prod_q \mathbb{Z}_q^\times$. Every class of the table is defined by congruences modulo an integer $M(\kappa)$ (for I3, modulo $4cdf$); we write $\text{Cl}(\kappa) \subset \widehat{\mathbb{Z}}$ for the set of $x \in \widehat{\mathbb{Z}}$ satisfying them. It is open and closed (*clopen*), and an integer n lies in $\text{Cl}(\kappa)$ exactly when it lies in the class. We call $\text{Cl}(\kappa)$ an *ET class*.

Definition 2.3. A point $x \in \widehat{\mathbb{Z}}^\times$ is *sterile* if it lies in no ET class. For a subfamily $\mathcal{K}$ of parameter triples, x is $\mathcal{K}$ -*sterile* if it lies in no $\text{Cl}(\kappa)$, $\kappa \in \mathcal{K}$.

Let $\mathcal{P}$ be an infinite set of primes and $\mathcal{P}'$ its set of accumulation points in $\widehat{\mathbb{Z}}$.

Lemma 2.4 (PROVED). *$\mathcal{P}'$ is a compact subset of $\widehat{\mathbb{Z}}^\times$. If $\Sigma \subset \widehat{\mathbb{Z}}^\times$ is clopen, defined modulo M (a union of reduced classes modulo M), and $\mathcal{P}(\Sigma)$ is the set of primes $p \nmid M$ with $p \bmod M$ in $\Sigma \bmod M$, then $\mathcal{P}(\Sigma)' = \Sigma$.*

Proof. If $x \in \mathcal{P}'$, then for each prime q and each k infinitely many $p \in \mathcal{P}$ satisfy $p \equiv x \pmod{q^k}$; all but one of them are prime to q , so $x_q \in \mathbb{Z}_q^\times$.

Derived sets in the compact metric space $\widehat{\mathbb{Z}}$ are closed, hence compact. For the second statement, a point of $\mathcal{P}(\Sigma)'$ is congruent modulo M to some element of $\mathcal{P}(\Sigma)$, so it lies in Σ . Conversely every neighbourhood $x + Q\widehat{\mathbb{Z}}$ ($M \mid Q$) of $x \in \Sigma$ is a reduced class modulo Q and contains infinitely many primes by Dirichlet's theorem; they lie in $\mathcal{P}(\Sigma)$. $\square$

The sets of primes studied below are all of the form $\mathcal{P}(\Sigma)$. The prime p itself is never a unit of $\mathbb{Z}_p$, which is why we work with accumulation points rather than with the primes as points of $\widehat{\mathbb{Z}}$.

Proposition 2.5 (PROVED). *Let $\mathcal{P}$ be an infinite set of primes.*

- (a) *If no point of $\mathcal{P}'$ is sterile, then finitely many ET classes contain all but finitely many $p \in \mathcal{P}$. For those p , Lemma 2.1 gives an explicit solution of $4/p = 1/x + 1/y + 1/z$.*
- (b) *If some $x \in \mathcal{P}'$ is sterile, then no finite set of ET classes contains all but finitely many $p \in \mathcal{P}$. Moreover no finite set of primitive residue classes, each solvable by polynomials, contains all but finitely many $p \in \mathcal{P}$.*

Hence all but finitely many elements of $\mathcal{P}$ lie in finitely many primitive classes solvable by polynomials if and only if $\mathcal{P}'$ contains no sterile point.

Proof. (a) The ET classes are open and cover the compact set $\mathcal{P}'$, so finitely many of them cover it; let U be their union. If $\mathcal{P} \setminus U$ were infinite, it would have an accumulation point, which lies in $\mathcal{P}' \subset U$; but U is open and misses $\mathcal{P} \setminus U$, a contradiction.

(b) Removing finitely many elements from $\mathcal{P}$ does not change $\mathcal{P}'$. If finitely many ET classes contain a cofinite subset of $\mathcal{P}$, their union is closed and therefore contains x , which contradicts sterility. Next let $C_1, \dots, C_s$ be primitive classes solvable by polynomials whose union contains a cofinite subset of $\mathcal{P}$. Viewed in $\widehat{\mathbb{Z}}$, their union is closed, so $x \in C_j$ for some j . By Cited Theorem 2.2 there are ET classes $K_1, \dots, K_t$ containing all large primes of C_j . Their union is closed and misses x , so some neighbourhood $V = x + Q\widehat{\mathbb{Z}} \subset C_j$ misses it. As x is a unit, V is a reduced class modulo Q , and by Dirichlet's theorem it contains infinitely many primes. These lie in C_j and in no K_i , a contradiction. The final equivalence combines (a), (b) and Lemma 2.1. $\square$

Remark 2.6 (Direction of the implications). The equivalence concerns the *existence* of a sterile point of $\mathcal{P}'$. For a *given* point x only one direction holds: if x is sterile there is no finite covering, but if x lies in some ET class another point of $\mathcal{P}'$ may still be sterile. An earlier internal draft of [12] said “equivalently” here; this was false and was corrected in review. A weaker statement is available for every explicit point: if $x \in \mathcal{P}'$ lies in no ET class of modulus at most Y , then every finite ET covering of a cofinite subset of $\mathcal{P}$ uses a class of modulus greater than Y . Indeed the classes of modulus $\leq Y$ in the covering miss a neighbourhood of x , which contains infinitely many elements of $\mathcal{P}$.

Remark 2.7 (Heights). The same compactness argument works for any subfamily $\mathcal{K}$ and any function $\kappa \mapsto \text{ht}(\kappa)$. Let $\mathcal{K}_X$ be the triples of height at most X and $U_X = \Sigma \setminus \bigcup_{\kappa \in \mathcal{K}_X} \text{Cl}(\kappa)$ for a compact Σ . The U_X are closed and decreasing. If each is nonempty, the finite intersection property gives a point of $\bigcap_X U_X$, which is $\mathcal{K}$ -sterile. So either some finite subfamily of bounded height covers Σ , or Σ has a $\mathcal{K}$ -sterile point. The set of $\mathcal{K}$ -sterile points of Σ is $\bigcap_X U_X$, a closed set.

3. SQUARE POINTS AND THE ODD-SQUARE OBSTRUCTION

Proposition 3.1 (PROVED; known). *Let $x \in \widehat{\mathbb{Z}}^\times$ be a square point: $x_q \in (\mathbb{Z}_q^\times)^2$ for every prime q . Then x is sterile.*

Proof. For $x = 1$ this is elementary: if $1 \in \text{Cl}(\kappa)$ then the integer $n = 1$ lies in the class, and Lemma 2.1 gives positive integers with $4 = 1/x + 1/y + 1/z \leq 3$. In general, suppose $x \in \text{Cl}(\kappa)$ with $\text{Cl}(\kappa)$ defined modulo M . For each $q \mid M$ the component x_q is the square of a unit, so by the Chinese remainder theorem there is an integer m prime to M with $m^2 \equiv x \pmod{M}$. The primitive class $m^2 \pmod{M}$ is then contained in the class of κ and is solvable by polynomials (Lemma 2.1). This contradicts the theorem of Mordell and Schinzel that a primitive class whose residue is a perfect square is not solvable by polynomials [5, 15], [2, p. 8]. $\square$

The last step is the classical obstruction; ET also derive it from their vanishing result at odd squares [2, Prop. 1.6], which goes back to Schinzel and Yamamoto [17]. The input is one quadratic character computation; Bright and Loughran [1] identify it with the Brauer–Manin obstruction of the Erdős–Straus surfaces.

Call a prime p *Mordell-hard* if $p \pmod{840}$ is a square, $p \pmod{840} \in \{1, 121, 169, 289, 361, 529\}$. Every primitive class modulo 840 that is not a square is solvable by polynomials (Mordell; see [2, p. 8]). The set $\Sigma_{\text{hard}} \subset \widehat{\mathbb{Z}}^\times$ of square classes modulo 840 contains the point 1, so by Propositions 2.5(b) and 3.1 the Mordell-hard primes admit no finite covering by polynomial identities. This is the classical form of the obstruction, and it is the reason why every finite covering statement below restricts the hard primes further by a *non-residue* condition at some additional prime r ; the resulting target sets contain no square point. Whether they contain other sterile points is the question of this note.

3.1. The procedure-level form. The project’s working note [8] proves a version of the odd-square obstruction for procedures. We only state it informally and refer to [8, §§1–2] and [7] for the model of computation. A *witness-producing program* takes a prime p , performs ring operations, floor divisions, eventual-sign comparisons, full factorisations and loops over the lists so produced, and outputs a solution or FAIL; it is *correct* if every solution it outputs is genuine, and *bounded* if the number of steps is bounded

independently of p . For $P \in \mathbb{Z}[X]$ a point $q^* \in \widehat{\mathbb{Z}}$ is *square-mimicking* for P if $P(q_\ell^*)$ is the square of a unit of $\mathbb{Z}_\ell$ for every prime ℓ .

Cited theorem 3.2 (Formal odd-square principle [8, Theorem C, Corollary C1]; CONDITIONAL on Hypothesis H). Let $P = uX + v$ be linear and q^* square-mimicking for P and nondegenerate in the sense of [8, §1.2]. If a correct witness-producing program has a defined and finite formal run at (P, q^*) , then it outputs FAIL at every admissible q (PROVED); under Schinzel’s Hypothesis H [16] for an explicit finite family there are infinitely many admissible q , and $p = P(q) \equiv 1 \pmod{24}$. In particular, under H for one explicit finite family, every correct bounded witness-producing program fails for infinitely many primes $p \equiv 1 \pmod{24}$.

A finite list of ET classes, each tested by a congruence and followed by the identity of Lemma 2.1, is such a bounded program, and its failure at the rational square-mimicking point 1 is Proposition 3.1 for $x = 1$, unconditionally. Cited Theorem 3.2 does not reprove the unconditional statement for general square points; it extends the obstruction from congruence coverings to all bounded procedures, at the price of Hypothesis H. *Novelty*. Theorem C is the procedure-level, H-conditional generalisation of known obstructions (Mordell and Schinzel on square classes, ET Prop. 1.6 and the remark on [2, p. 6] that “a finite set of covering congruence strategies” must fail, Yamamoto, Bright–Loughran); its unconditional instances are known in substance. The project’s novelty audit rates it as a partial novelty only.

Remark 3.3 (ASSESSMENT). The target sets of §§??–?? require x_r to be a non-square for some prime r , so they contain no point that is square-mimicking for $P = X$, and neither Proposition 3.1 nor Cited Theorem 3.2 applies to them. The candidate sterile points below (the sign point $\hat{x}_9$, the point x^* for $r = 13$, the 17-generic line) are therefore not explained by the odd-square obstruction. If they are sterile, the reason is a different, rigidity type of obstruction (§??); this is a heuristic reading, not a theorem.

ACKNOWLEDGEMENTS AND PROVENANCE

PROVENANCE-PLACEHOLDER

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- [9] Companion note in this repository, [POINTWISE_TYPEI.md](#); internally reviewed in [reviews/pointwise-typei-review.md](#).
- [10] Companion note in this repository, [POINTWISE_TYPEI2.md](#); internally reviewed in [reviews/pointwise-typei2-review.md](#).
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